

polygonal region P . Therefore, G also extends to a continuous function from $\overline{P}$ to $\overline{\mathbb{D}}$. It suffices to now prove that the extensions of F and G are inverses of each other. If $z \in \partial\mathbb{D}$ and $\{z_k\}$ is a sequence in the disc that converges to z , then $G(F(z_k)) = z_k$, so after taking the limit and using the fact that F is continuous, we conclude that $G(F(z)) = z$ for all $z \in \overline{\mathbb{D}}$. Similarly, $F(G(w)) = w$ for all $w \in \overline{P}$, and the theorem is proved.

The circle of ideas used in this proof can be used to prove more general theorems on the boundary continuity of conformal maps. See Exercise 18 and Problem 6 below.

4.4 The mapping formula

Suppose P is a polygonal region bounded by a polygon $\mathbf{p}$ whose vertices are ordered consecutively $a_1, a_2, \dots, a_n$, and with $n \geq 3$. We denote by $\pi\alpha_k$ the interior angle at a_k , and define the exterior angle $\pi\beta_k$ by $\alpha_k + \beta_k = 1$. A simple geometric argument provides $\sum_{k=1}^n \beta_k = 2$.

We shall consider conformal mappings of the half-plane $\mathbb{H}$ to P , and make use of the results of the previous section regarding conformal maps from the disc $\mathbb{D}$ to P . The standard correspondences $w = (i - z)/(i + z)$, $z = i(1 - w)/(1 + w)$ allows us to go back and forth between $z \in \mathbb{H}$ and $w \in \mathbb{D}$. Notice that the boundary point $w = -1$ of the circle corresponds to the point at infinity on the line, and so the conformal map of $\mathbb{H}$ to $\mathbb{D}$ extends to a continuous bijection of the boundary of $\mathbb{H}$, which for the purpose of this discussion includes the point at infinity.

Let F be a conformal map from $\mathbb{H}$ to P . (Its existence is guaranteed by the Riemann mapping theorem and the previous discussion.) We assume first that none of the vertices of $\mathbf{p}$ correspond to the point at infinity. Therefore, there are real numbers $A_1, A_2, \dots, A_n$ so that $F(A_k) = a_k$ for all k . Since F is continuous and injective, and the vertices are numbered consecutively, we may conclude that the A_k 's are in either increasing or decreasing order. After relabeling the vertices a_k and the points A_k , we may assume that $A_1 < A_2 < \dots < A_n$. These points divide the real line into $n - 1$ segments $[A_k, A_{k+1}]$, $1 \leq k \leq n - 1$, and the segment that consists of the join of the two half-segments $(-\infty, A_1] \cup [A_n, \infty)$. These are mapped bijectively onto the corresponding sides of the polygon, that is, the segments $[a_k, a_{k+1}]$, $1 \leq k \leq n - 1$, and $[a_n, a_1]$ (see Figure 8).

Theorem 4.6 *There exist complex numbers c_1 and c_2 so that the conformal map F of $\mathbb{H}$ to P is given by*

$$F(z) = c_1 S(z) + c_2$$

where S is the Schwarz-Christoffel integral introduced in Section 4.2.

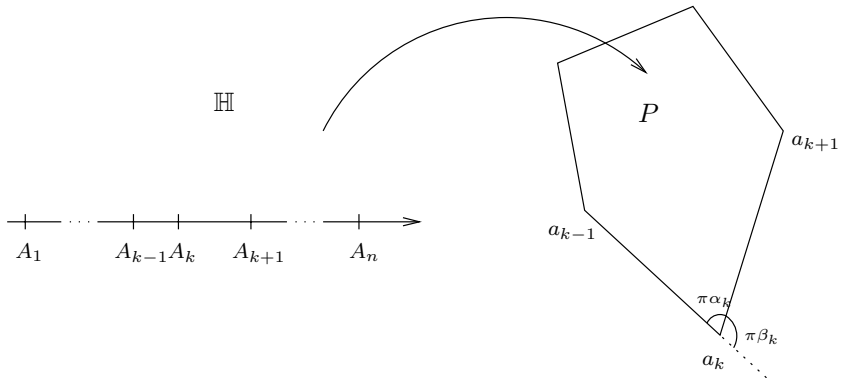


Figure 8. The mapping F

Proof. We first consider z in the upper half-plane lying above the two adjacent segments $[A_{k-1}, A_k]$ and $[A_k, A_{k+1}]$, where $1 < k < n$. We note that F maps these two segments to two segments that intersect at $a_k = F(A_k)$ at an angle $\pi\alpha_k$.

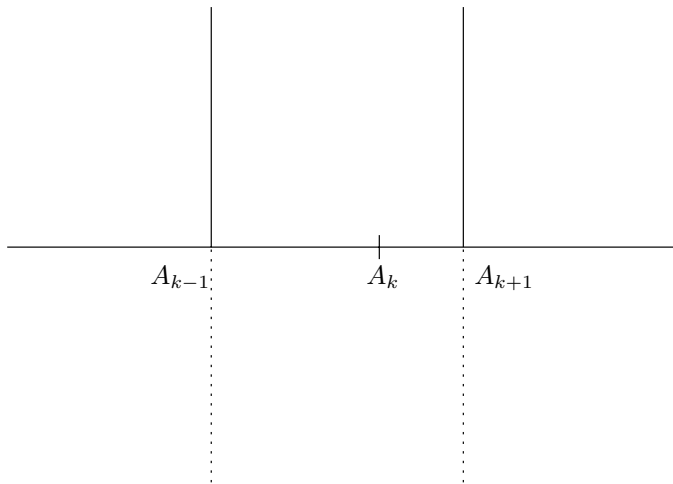
By choosing a branch of the logarithm we can in turn define

$$h_k(z) = (F(z) - a_k)^{1/\alpha_k}$$

for all z in the half-strip in the upper half-plane bounded by the lines $\operatorname{Re}(z) = A_{k-1}$ and $\operatorname{Re}(z) = A_{k+1}$. Since F continues to the boundary of $\mathbb{H}$, the map h_k is actually continuous up to the segment (A_{k-1}, A_{k+1}) on the real line. By construction h_k will map the segment $[A_{k-1}, A_{k+1}]$ to a (straight) segment L_k in the complex plane, with A_k mapped to 0. We may therefore apply the Schwarz reflection principle to see that h_k is analytically continuable to a holomorphic function in the two-way infinite strip $A_{k-1} < \operatorname{Re}(z) < A_{k+1}$ (see Figure 9). We claim that h'_k never vanishes in that strip. First, if z belongs to the open upper half-strip, then

$$\frac{F'(z)}{F(z) - F(A_k)} = \alpha_k \frac{h'_k(z)}{h_k(z)},$$

and since F is conformal, we have $F'(z) \neq 0$ so $h'_k(z) \neq 0$ (Proposition 1.1). By reflection, this also holds in the lower half-strip, and it remains to investigate points on the segment (A_{k-1}, A_{k+1}) . If $A_{k-1} < x < A_{k+1}$, we note that the image under h_k of a small half-disc centered at x and contained in $\mathbb{H}$ lies on one side of the straight line segment

**Figure 9.** Schwarz reflection

L_k . Since h_k is injective up to L_k (because F is) the symmetry in the Schwarz reflection principle guarantees that h_k is injective in the whole disc centered at x , whence $h'_k(x) \neq 0$, whence $h'_k(z) \neq 0$ for all z in the strip $A_{k-1} < \operatorname{Re}(z) < A_{k+1}$.

Now because $F' = \alpha_k h_k^{-\beta_k} h'_k$ and $F'' = -\beta_k \alpha_k h_k^{-\beta_k-1} (h'_k)^2 + \alpha_k h_k^{-\beta_k} h''_k$, the fact that $h'_k(z) \neq 0$ implies that

$$\frac{F''(z)}{F'(z)} = \frac{-\beta_k}{z - A_k} + E_k(z),$$

where E_k is holomorphic in the strip $A_{k-1} < \operatorname{Re}(z) < A_{k+1}$. A similar result holds for $k = 1$ and $k = n$, namely

$$\frac{F''(z)}{F'(z)} = -\frac{\beta_1}{z - A_1} + E_1,$$

where E_1 is holomorphic in the strip $-\infty < \operatorname{Re}(z) < A_2$, and

$$\frac{F''(z)}{F'(z)} = -\frac{\beta_n}{z - A_n} + E_n,$$

where E_n is holomorphic in the strip $A_{n-1} < \operatorname{Re}(z) < \infty$. Finally, another application of the reflection principle shows that F is continuable in the exterior of a disc $|z| \leq R$, for large R (say $R > \max_{1 \leq k \leq n} |A_k|$). Indeed, we may continue F across the union of the segments

$(-\infty, A_1) \cup (A_n, \infty)$ since their image under F is a straight line segment and Schwarz reflection applies. The fact that F maps the upper half-plane to a bounded region shows that the analytic continuation of F outside a large disc is also bounded, and hence holomorphic at infinity. Thus F''/F' is holomorphic at infinity and we claim that it goes to 0 as $|z| \rightarrow \infty$. Indeed, we may expand F at $z = \infty$ as

$$F(z) = c_0 + \frac{c_1}{z} + \frac{c_2}{z^2} + \cdots.$$

This after differentiation shows that F''/F' decays like $1/z$ as $|z|$ becomes large, and proves our claim.

Altogether then, because the various strips overlap and cover the entire complex plane,

$$\frac{F''(z)}{F'(z)} + \sum_{k=1}^n \frac{\beta_k}{z - A_k}$$

is holomorphic in the entire plane and vanishes at infinity; thus, by Liouville's theorem it is zero. Hence

$$\frac{F''(z)}{F'(z)} = - \sum_{k=1}^n \frac{\beta_k}{z - A_k}.$$

From this we contend that $F'(z) = c(z - A_1)^{-\beta_1} \cdots (z - A_n)^{-\beta_n}$. Indeed, denoting this product by $Q(z)$, we have

$$\frac{Q'(z)}{Q(z)} = - \sum_{k=1}^n \frac{\beta_k}{z - A_k}.$$

Therefore

$$\frac{d}{dz} \left(\frac{F'(z)}{Q(z)} \right) = 0,$$

which proves the contention. A final integration yields the theorem.

We may now withdraw the hypothesis we made at the beginning that F did not map the point at infinity to a vertex of P , and obtain a formula for that case as well.

Theorem 4.7 *If F is a conformal map from the upper half-plane to the polygonal region P and maps the points $A_1, \dots, A_{n-1}, \infty$ to the vertices of $\mathfrak{p}$, then there exist constants C_1 and C_2 such that*

$$F(z) = C_1 \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_{n-1})^{\beta_{n-1}}} + C_2.$$

In other words, the formula is obtained by deleting the last term in the Schwarz-Christoffel integral (5).

Proof. After a preliminary translation, we may assume that $A_j \neq 0$ for $j = 1, \dots, n-1$. Choose a point $A_n^* > 0$ on the real line, and consider the fractional linear map defined by

$$\Phi(z) = A_n^* - \frac{1}{z}.$$

Then Φ is an automorphism of the upper half-plane. Let $A_k^* = \Phi(A_k)$ for $k = 1, \dots, n-1$, and note that $A_n^* = \Phi(\infty)$. Then

$$(F \circ \Phi^{-1})(A_k^*) = a_k \quad \text{for all } k = 1, 2, \dots, n.$$

We can now apply the Schwarz-Christoffel formula just proved to find that

$$(F \circ \Phi^{-1})(z') = C_1 \int_0^{z'} \frac{d\zeta}{(\zeta - A_1^*)^{\beta_1} \cdots (\zeta - A_n^*)^{\beta_n}} + C_2.$$

The change of variables $\zeta = \Phi(w)$ satisfies $d\zeta = dw/w^2$, and since we can write $2 = \beta_1 + \cdots + \beta_n$, we obtain

$$\begin{aligned} (F \circ \Phi^{-1})(z') &= C_1 \int_0^{\Phi^{-1}(z')} \frac{dw}{(w(A_n^* - A_1^*) - 1)^{\beta_1} \cdots (w(A_n^* - A_{n-1}^*) - 1)^{\beta_{n-1}}} + C_2' \\ &= C_1' \int_0^{\Phi^{-1}(z')} \frac{dw}{(w - 1/(A_n^* - A_1^*))^{\beta_1} \cdots (w - 1/(A_n^* - A_{n-1}^*))^{\beta_{n-1}}} + C_2'. \end{aligned}$$

Finally, we note that $1/(A_n^* - A_k^*) = A_k$ and set $\Phi^{-1}(z') = z$ in the above equation to conclude that

$$F(z) = C_1' \int_0^z \frac{dw}{(w - A_1)^{\beta_1} \cdots (w - A_{n-1})^{\beta_{n-1}}} + C_2',$$

as was to be shown.

4.5 Return to elliptic integrals

We consider again the elliptic integral

$$I(z) = \int_0^z \frac{d\zeta}{[(1 - \zeta^2)(1 - k^2\zeta^2)]^{1/2}} \quad \text{with } 0 < k < 1,$$